

Kuldeep Lodhi

Senior Software Engineer

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Professional Experience

Reliance Jio Platforms, Software Development Engineer - 2

Dec 2023 – Present
Navi Mumbai, India

- Led frontend development for an enterprise inventory platform serving **1,300+ stores** across 6 brands managing **INR 50+ Crores** in inventory with **6,000+ daily active users**
- Built **30+ reusable TypeScript components** and standardized Redux architecture, reducing frontend duplication by 30%
- Improved dashboard performance by **45%** using memoization and virtualization, enabling real-time operational dashboards
- Mentored a **3-member frontend team**, conducting code reviews and driving Agile delivery
- Implemented **Docker containerization and CI/CD pipelines** via Azure DevOps, enabling frequent production releases

Algonauts Advisory, Senior Software Developer

Nov 2021 – Dec 2023
Mumbai, India

- Architected a full-stack **trading simulator platform** using Svelte.js serving **400+ daily users** and processing **8K+ trades/day**
- Collaborated directly with founders to design product architecture and trading workflows
- Implemented an ML-based portfolio recommendation system increasing engagement by **25%**
- Scaled the engineering team from 1 to 3 developers while maintaining **99.5% uptime**

Ugam Solutions, Senior Analyst

Jun 2019 – Nov 2021
Thane, India

- Delivered **15+ enterprise analytics projects** for CDC, University of Chicago, Google Screenwise, and IPSOS
- Developed JavaScript libraries and dashboards reducing report generation time by **40%**

Key Projects

JVA Retail Video Analytics Platform | React, TypeScript, Konva, Kafka

- Built dashboards for an **AI-powered video analytics platform** converting camera streams into retail operational insights
- Implemented modules for **traffic analytics, queue monitoring, customer demographics, and store heatmaps**
- Developed a **store planner interface** for floor-plan mapping and camera placement enabling zone-based visitor analytics
- Integrated a **Kafka streaming pipeline** where camera feeds are published to Kafka and consumed by multiple analytics pipelines processing feeds against defined **Regions of Interest (ROIs)**
- Implemented **Server-Sent Events (SSE)** enabling real-time **alerts and event triggers** for queue thresholds, congestion, and system events

AnyTrac RFID Platform | React, TypeScript, Redux, Node.js, Kafka

- Implemented unified RFID-based inventory tracking enabling **50-item carton scanning in 1 second vs 5 minutes manually**

- Improved inventory accuracy from **75% to 98%** and reduced shrinkage by **40%**
- Designed an event-driven architecture where **Product SOH (Stock on Hand)** and tracking updates are pushed to **Kafka queues** and consumed by downstream services
- Supported **Store Portal, Cluster Portal, and Self-Checkout systems** across **1,000+ retail locations**

Reliance Jewels Inventory Platform | React, TypeScript, Redux

- Built an enterprise inventory management platform for **Reliance Jewels stores** supporting daily stock reconciliation and store operations
- Developed dashboards for **Stock-on-Hand (SOH)**, inward/outward tracking, and real-time stock visibility across store clusters
- Enabled store managers to monitor inventory movements, reducing reconciliation time and improving operational accuracy
- Implemented performance optimizations and reusable UI components improving dashboard responsiveness for large datasets

Reliance Foundation Scholarship Portal | React, Node.js, MongoDB

- Built a scholarship management system handling **50,000+ annual applications** with OCR-based document ingestion
- Migrated a legacy K2 workflow system to a **MERN architecture**, improving scalability by **200%**
- Developed an interview management portal supporting **50+ concurrent interview sessions** with automated evaluation workflows

Technical Skills

Frontend: React, TypeScript, Next.js, Redux, Svelte, Konva, Chart.js, ApexCharts, D3.js

UI Libraries: Material-UI, Tailwind CSS, Bootstrap

Backend: Node.js, Express, Python, Django, REST APIs, Microservices

Streaming & Messaging: Apache Kafka, Event-driven Architecture

Database: MongoDB, PostgreSQL, MySQL, Redis

DevOps & Cloud: Docker, CI/CD, Azure DevOps, AWS, Git

Frontend Architecture: Component Design Patterns, State Management, Code Splitting, Lazy Loading

Performance Optimization: Memoization, Virtualization, Bundle Optimization

System Design: Scalable Systems, Performance Optimization, Distributed Systems

Security: Authentication, Authorization

Leadership & Collaboration: Agile/Scrum (Sprint Planning, Story Estimation, Standups, Retrospectives), Team Mentoring (3+ Engineers), Code Reviews (30+/Sprint), Cross-functional Collaboration, Stakeholder Management

Education

Bachelor of Engineering in Computer Engineering
Vidyalankar Institute of Technology, Mumbai

2015 – 2019
CGPI: 7.95/10